

Siyuan Mei

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RESEARCH INTERESTS

Medical AI: reconstruction, synthesis, segmentation, and registration.

Computer vision: diffusion and flow-matching models, generative modeling, representation learning, and dense prediction.

LLMs & software engineering: LLM-assisted development, reproducible research tooling, and agentic systems.

EDUCATION

FAU Erlangen-Nürnberg 2023–2027 (expected)

Ph.D. in Computer Science; Supervisor: Prof. Andreas Maier.

Peking University Health Science Center May–Oct. 2026

Visiting Scholar, Artificial Intelligence for Medical Imaging (AIMI) Lab; Supervisor: Prof. Yixing Huang.

FAU Erlangen-Nürnberg 2020–2023

M.Sc. in Medical Engineering; thesis grade: 1.0.

Zhejiang University of Science and Technology 2016–2020

B.Eng. in Automation; Sino-German 2+3 Program; honored graduate scholarship.

PROFESSIONAL EXPERIENCE

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) 2023–2027

Ph.D. Student, Pattern Recognition Lab; Supervisor: Prof. Andreas Maier

- Fully funded through the ERC project *Advancing osteoporosis medicine by observing bone microstructure and remodelling using a 4D nanoscope*; developed **BigReg**, an efficient registration pipeline for high-resolution X-ray microscopy and light-sheet fluorescence microscopy.
- Conduct research in medical AI and computer vision, adapting foundation models and developing medical image generation methods with GANs, diffusion models, and flow matching.
- Teach *Flat-panel CT Reconstruction* and contribute lecture slides for *Vibe Coding: Software Engineering with LLMs*, a Springer book project.
- Outcomes:** first-author work in Pattern Recognition and MICCAI; a Featured Paper in Journal of Medical Imaging; and an Oral Presentation at MICCAI 2026.

Peking University Health Science Center, AIMI Lab May–Oct. 2026

Visiting Scholar; Supervisor: Prof. Yixing Huang

Artificial Intelligence for Medical Imaging Lab, Institute of Medical Technology, 240 Pathology Building, No. 38 Xueyuan Road, Haidian District, Beijing, China.

- Participated in the MICCAI BIC-MAC Challenge and trained a multimodal-to-CT generative model for PET attenuation correction; achieved first place on the validation-set CT metric.
- Built practical knowledge of clinical CT applications and imaging workflows; visited the ICT Center in Chongqing and attended MISC 2026 at Tsinghua University.

SELECTED PUBLICATIONS

J = Journal, C = Conference, M = Manuscript

- [C.1] Siyuan Mei et al. **Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation.** MICCAI 2026; **Early Accept; Oral Presentation**; first author.
- [C.2] Mengchen Fan et al., including Siyuan Mei. **Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation.** MICCAI 2026; **Early Accept; Spotlight Presentation.**
- [C.3] Daiqi Liu et al., including Siyuan Mei. **Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI.** MICCAI 2026; **Oral Presentation.**
- [C.4] Siyuan Mei et al. **WING: A Window-Prior-Based Generative Network with Gated Inception for Cross-Modality CT Synthesis.** MICCAI MIART Workshop 2026; accepted workshop paper; first author.
- [J.1] Siyuan Mei et al. **Vision Transformer Hook for Dense Predictions.** Pattern Recognition 179 (2026), 113818; first author.

- [J.2] Siyuan Mei et al. **BigReg: An Efficient Registration Pipeline for High-Resolution X-Ray and Light-Sheet Fluorescence Microscopy**. Journal of Medical Imaging 12(5), 054004 (2025); [Featured Paper](#).
- [C.5] Siyuan Mei, F. Fan, A. Maier. **Metal Inpainting in CBCT Projections Using Score-based Generative Model**. IEEE ISBI 2023; first author.
- [C.6] Siyuan Mei (co-author). **TCCT: Trajectory-Conditioned CBCT Reconstruction for Sinusoidal Non-Circular Orbits with a Fourier Neural Operator**. MICCAI 2026; Poster.
- [J.3] Y. Sun et al., including Siyuan Mei. **Filter2Noise: A Framework for Interpretable and Zero-Shot Low-Dose CT Image Denoising**. Journal of Medical Imaging 13(2), 024004 (2026).
- [J.4] C. Ye et al., including Siyuan Mei. **Robustness and Stability Analysis of Differentiable Shift-Variant FBP for Cone-Beam CT under Challenging Acquisition Settings**. Machine Learning for Biomedical Imaging (2026).
- [J.5] M. Thies et al., including Siyuan Mei. **A Gradient-Based Approach to Fast and Accurate Head Motion Compensation in Cone-Beam CT**. IEEE Transactions on Medical Imaging (2024).
- [J.6] C. Ye et al., including Siyuan Mei. **DRACO: Differentiable Reconstruction for Arbitrary CBCT Orbits**. Physics in Medicine and Biology 70, 075005 (2025).
- [C.7] M. Gu et al., including Siyuan Mei. **Unsupervised Domain Adaptation Using Soft-Labeled Contrastive Learning with Reversed Monte Carlo Method for Cardiac Image Segmentation**. MICCAI 2024.
- [C.8] L. Pfaff et al., including Siyuan Mei. **No-New-Denoiser: A Critical Analysis of Diffusion Models for Medical Image Denoising**. MICCAI 2024.
- [M.1] Zirong Li, Siyuan Mei et al. **Generative Drifting for Conditional Medical Image Generation**. arXiv preprint (2026).
- [M.2] Siyuan Mei, Y. Xia, F. Fan. **GANeXt: A Fully ConvNeXt-Enhanced Generative Adversarial Network for MRI- and CBCT-to-CT Synthesis**. arXiv preprint (2025).

HONORS AND AWARDS

- **1st place, Task 2** and **4th place, Task 1**, MICCAI SynthRAD2025 Challenge, 2025.
- **Two merit-based scholarships** during M.Sc. studies at FAU Erlangen-Nürnberg.
- Fully funded Ph.D. through the ERC 4D Nanoscope project, FAU, 2023–2027.
- Honored graduate scholarship, Zhejiang University of Science and Technology, 2020.

TEACHING

Flat-panel CT Reconstruction Summer semesters 2024–2026.	FAU
Vibe Coding: Software Engineering with LLMs Summer semester 2026; lecture slides accompanying a Springer book.	FAU

ACADEMIC SERVICES

Journal reviewer: Pattern Recognition; Journal of Machine Learning for Biomedical Imaging (MELBA).
Technical editor: Journal of Medical Imaging.
Conference reviewer: MICCAI.

ADDITIONAL INFORMATION

Languages: Chinese (native), English (professional), German (C1 certificate).
Interests: fitness and tennis (level 2.5). **MBTI (self-reported):** ENTJ-A.
Career conversations: I am open to research conversations, collaborations, and job-related discussions in medical imaging, computer vision, and LLM-assisted software engineering; I appreciate hearing about suitable opportunities.
Open-source and teaching resources: [WING](#), [JiR](#), [ViT-Hook](#), and [Vibe Coding lecture slides](#) accompanying a Springer book.